Paired t-Test

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Introduction

When each unit contributes two measurements (pre and post, left and right, case and matched control), the observations are paired and the usual independent-samples t-test is not valid. The **paired t-test** analyses the within-unit differences, restoring the independence assumption and typically improving power by cancelling between-unit variability.

Prerequisites

One-sample t-test, t-distribution.

Theory

Let X_i and Y_i be paired measurements on unit i for $i = 1, \ldots, n$. Define differences $D_i = X_i - Y_i$. Under the assumption that D_i are iid from $\mathcal{N}(\mu_D, \sigma_D^2)$,

$$T = \frac{\bar{D}}{s_D/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0 : \mu_D = 0.$$

The paired t-test is the one-sample t-test on the differences. The n is the number of **pairs**, not the total number of measurements.

Effect size: Cohen's $d_z = \bar{D}/s_D$ (standardised mean difference on the difference scores). This is different from the two-group d and should not be conflated.

Assumptions

- The differences are approximately normal (via Shapiro-Wilk on D_i).
- Pairs are independent of other pairs.

The raw X_i and Y_i can be non-normal; only the differences need to be.

R Implementation

```r
library(effectsize); library(broom)
set.seed(2026)

# Simulated pre-post blood pressure after a 12-week intervention
n <- 30
pre  <- rnorm(n, mean = 138, sd = 12)
post <- pre + rnorm(n, mean = -6, sd = 5)

# Paired t-test
t.test(pre, post, paired = TRUE) |> tidy()

# Equivalent: one-sample t-test on the differences
```

```r
t.test(pre - post, mu = 0) |> tidy()

# Effect size
cohens_d(pre, post, paired = TRUE)

# Assumption check on differences
shapiro.test(pre - post)
```

Output & Results

```
# A tibble: 1 x 8
  estimate statistic p.value parameter conf.low conf.high method        alternative
     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>         <chr>
1     6.04      6.88 <0.001         29     4.25      7.84 Paired t-... two.sided

Cohen's d (paired) |         95% CI
-------------------------------------
1.26               | [0.78, 1.74]
```

The post-intervention measurement is 6.0 mmHg lower on average; $t(29) = 6.88$, $p < 0.001$, large paired effect.

Interpretation

In a manuscript: "Blood pressure decreased by 6.0 mmHg on average from baseline to week 12 (paired t(29) = 6.88, p < 0.001, Cohen's d_z = 1.26, 95 % CI 4.3 to 7.8 mmHg)."

Practical Tips

- Paired design usually has more power than independent-groups design at the same total sample size.
- Check normality of the differences, not the raw measurements.
- For non-normal differences, use Wilcoxon signed-rank.
- Pair variables correctly: subjects must be in the same order in both vectors, or use a data frame.
- Report Cohen's d_z, not pooled-SD two-group d, for paired designs.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests